

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed biodegradable material formulations for **direct laser writing** of porous electrodes, integrating flexible and transparent features sourced from natural materials.
- Designed and implemented methodologies for universal **Laser Induced Graphene (LIG) fabrication** on biopolymers, producing electrode materials suitable for electrochemical sensing applications.
- Integrated fabricated electrodes with green sensing active layers to enable the detection of **biomarkers in e-textiles**, supporting the EU-funded e-FLOWER project.
- Collaborated with cross-departmental colleagues and e-FLOWER partners to develop novel integrated sensing **e-textile platforms** with reduced environmental footprints.
- Managed production environments for enterprise applications, ensuring SLA adherence and responding to P1/P2 incidents using **Splunk** and **Dynatrace**, reducing critical alert response times by approximately 15%.
- Automated data extraction and reporting tasks using **Python scripting** across PostgreSQL and Oracle databases, decreasing manual effort for weekly operational reports by ~8 hours.

SKILLS

Materials Science & Fabrication – Laser Induced Graphene (LIG), Direct Laser Writing (DLW), Biodegradable Materials, Nanoscale Materials Synthesis, Electrode Fabrication, Biomarker Detection, e-Textiles

Chemistry & Electrochemistry – Material Formulation, Electrochemical Sensing, Green Chemistry, Active Layer Integration

Software Engineering – Java, Spring Boot, Python, REST APIs, Docker, Linux, Git, Jenkins, Jira, Confluence

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch)

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, SQL (PostgreSQL, MySQL, Oracle), R/RStudio, pandas, NumPy, scikit-learn

Project Management & Compliance – Agile/Scrum, Health and Safety Standards, Quality Management Systems, GDPR Awareness, Regulatory Compliance

PROJECTS

Development of LIG Electrodes for Smart Textiles

- Engineered novel biodegradable polymer formulations for **direct laser writing**, optimizing for porosity and conductivity in e-textile applications.
- Developed and validated fabrication processes for **Laser Induced Graphene (LIG) electrodes** on flexible substrates, achieving consistent surface morphology and electrical properties across multiple batches.
- Integrated fabricated LIG electrodes with active sensing layers, demonstrating proof-of-concept for wearable biosensors within a simulated e-textile environment.

Automated Transaction Data Processing Pipeline

- Built a **Python-based ETL pipeline** to ingest and transform transaction data from PostgreSQL and Oracle databases, processing approximately 5,000 records daily.
- Developed **scheduled reporting** scripts that generated daily and weekly summaries, reducing manual data compilation time for the operations team by 6 hours per week.
- Implemented data validation checks within the pipeline to ensure data integrity and accuracy before loading into the financial-services dashboard.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)